ISBN: 978-3-95450-263-9 ISSN: 2673-5520 doi: 10.18429/JACoW-MEDSI2025-WEOA02

DESIGNING AND FINE TUNING CRYO-COOLED SILICON MONO-CROMATOR CRYSTALS TO MINIMIZE OPTICAL DISTORTIONS CAUSED BY PHOTON-BEAM HEATING

P. Sanchez Navarro[*,1,2], S. G. Alcock[1], A. Peach[1]
[1]Diamond Light Source Ltd., Didcot, UK
[2]MAX IV Laboratory, Lund, Sweden

Abstract

Slope errors on X-ray optics create distortions in the reflected or diffracted X-ray wavefront and a reduction in energy resolution. This study addresses this challenge by demonstrating a precise and adaptable method for tuning the geometry of liquid nitrogen (LN2)-cooled silicon crystals, with the goal of achieving zero slope errors under specified input power conditions. The findings reveal that an optimal temperature exists, minimizing thermal distortion and slope errors at the X-ray beam footprint. By establishing a straightforward engineering approach to achieve this temperature, the study provides a practical solution for manipulating silicon crystal geometry. This technique ensures minimal slope errors across a broad energy spectrum, enhancing beamline performance and energy resolution. This work overcomes a long-standing limitation in particle accelerator beamlines, where conventional approaches relied on extensive cooling to mitigate thermal effects. The proposed methodology not only improves operational efficiency but also offers a versatile tool for finetuning crystal behaviour in response to varying energy demands.

INTRODUCTION

Crystals with various geometries are used to select the energy bandwidth of intense photon beams from synchrotron and X-ray free-electron laser (XFEL) sources. The performance of monochromator crystals is reduced when diffraction planes are distorted by clamping strain or thermal expansion from heat loads. Slope errors (deviation from the ideal profile) distort the diffracted wavefront and degrade resolution.

Optical quality is a limiting factor for most beamlines, pushing the limits of commercially available solutions. Facilities such as ESRF, SLAC, and Diamond Light Source have shown how thermal deformation in cryogenically cooled silicon crystals impacts double-crystal monochromator performance [1, 2]. With the higher brightness and coherence of new accelerators such as HEPS and Diamond-II, robust optics are needed to manage heat-induced deformation [3].

Promising developments include channel-cut diamond crystals and advanced cooling schemes. Direct-cooling designs are particularly attractive for high-heat-load beamlines with wigglers. Previous tests on an XDS Oxford (formerly FMB Oxford) direct LN2-cooled crystal for the DLS I20 Spectroscopy Beamline confirmed superior performance of Si(111) compared with indirect cooling. However, for Si(311), additional design work was needed to meet the <0.5 µrad slope error requirement across 7–35 keV with zero clamping strain.

* pablo.sanchez_navarro@maxiv.lu.se

METHOD

At approximately –150 °C, silicon's coefficient of thermal expansion (CTE) is zero (see Fig. 1). Cooling further reverses the sign, peaking at –200 °C before returning to zero at –250 °C. Thus, there exists a temperature "sweet spot" where the beam footprint becomes flat, yielding zero slope error (see Fig. 2).

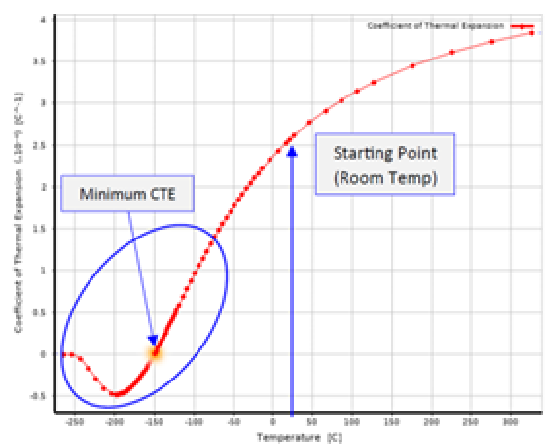

Figure 1: Thermal expansion coefficient of silicon vs. temperature from -250 °C up to 300 °C.

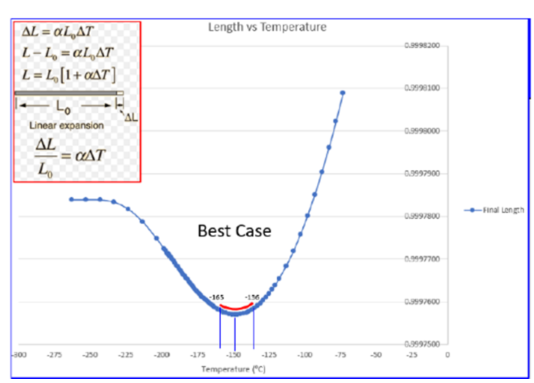

Figure 2: Linear expansion equation for a simple beam using the silicon thermal expansion coefficient chart.

ISBN: 978-3-95450-263-9 ISSN: 2673-5520 doi: 10.18429/JACoW-MEDSI2025-WEOA02

This behavior allows tailoring of crystal geometry to control footprint temperature. As Bragg angle varies, footprint size and power density change, but geometry can be adjusted to minimize slope errors at a target energy or compromise across a range. The design consists of a single solid silicon piece. Internal cooling channels were narrowed to increase flow velocity and heat transfer. Through cutouts separate hot and cold regions, raising the footprint temperature to the optimum while maintaining low channel-wall temperatures to prevent LN2 boiling. In Figs. 3 and 4, the thin boundary at –150 °C (CTE=0) outlines the footprint. The maximum temperature is –135.9 °C, meaning only the boundary has zero expansion. Inside, CTE is positive; outside, negative. Two optimized geometries were developed for Si(111) and Si(311), tailored for different input powers. Cutouts prevent LN2 boiling, while leg thickness and cold–hot region spacing control footprint temperature. Increased footprint distance from clamping also reduces distortion sensitivity.

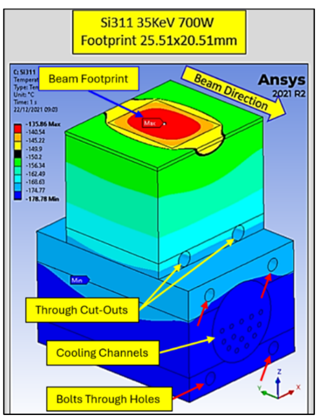

Figure 3: Optimized silicon-311 crystal design.

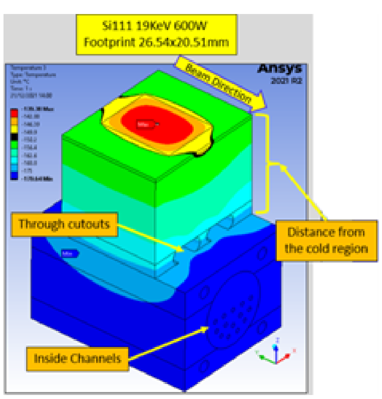

Figure 4: Optimized silicon-111 crystal design.

ANALYSIS VS. EXPERIMENTS

Experimental setup (see Fig. 5): The silicon assembly was tested in vacuum with PT100 sensors for comparison with FEA simulations. Heat input was applied by a surface heater, while LN2 circulated through a closed loop (see Fig. 6).

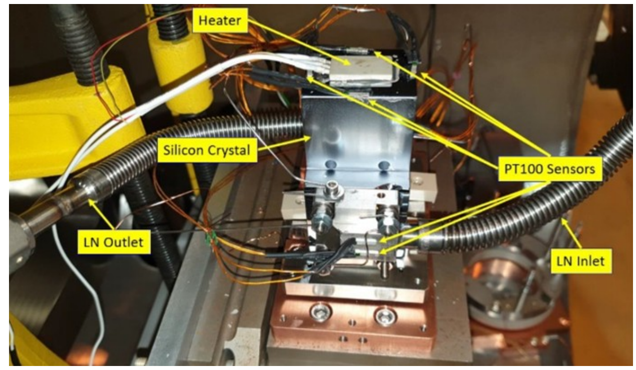

Figure 5: Experimental setup.

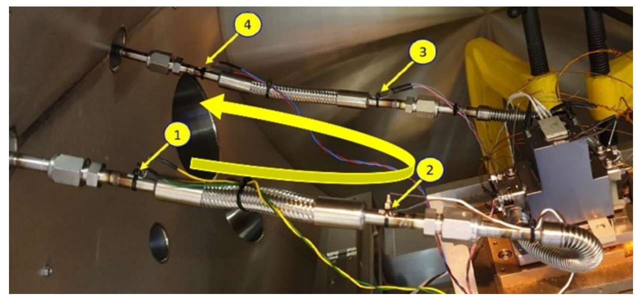

Figure 6: PT100 sensors at the cooling pipes.

PT100 sensors lose accuracy at cryogenic temperatures (see Fig. 7), and cryocooler sensors are affected by LN2 fluctuations. Therefore, the average inlet temperature (–194 °C from sensors 1 and 2) was used in FEA analysis (see Fig. 8).

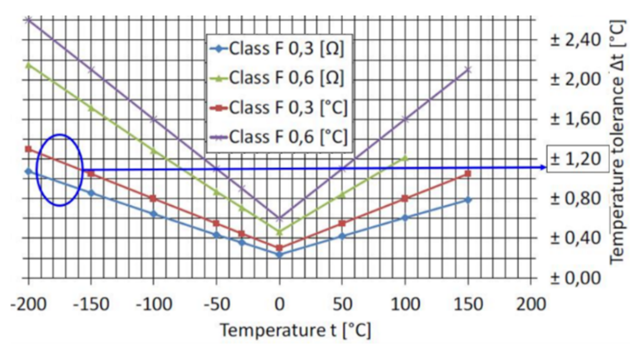

Figure 7: PT100 sensors tolerance vs. temperature.

Simulations closely matched 3 of 4 sensor readings, all within tolerance. A Micro Epsilon CS1 capacitive sensor measured vibration to validate FEA. Both simulation and experiment showed boiling onset at <4 L/min under 7 bar and increased boiling at 4 bar (see Fig. 9). A Fizeau laser interferometer measured optical surface changes from coolingend-cap clamping (see Figs. 10 and 12). Results compared well with FEA (see Fig. 11), showing strong agreement in vertical PV (4.718 nm vs. 4.699 nm) and reasonable agreement in horizontal PV (1.29 nm vs. 1.7 nm).

Minor asymmetry in experiments was attributed to uneven bolt forces and silicon anisotropy. Table 1 summarizes

ISBN: 978-3-95450-263-9 ISSN: 2673-5520 doi: 10.18429/JACoW-MEDSI2025-WEOA02

the FEA predictions of slope errors for Si(111) and Si(311) across input powers and flow rates showed promising results, with optimization targeted at 19 keV and 35 keV, respectively, while maintaining acceptable performance at other conditions.

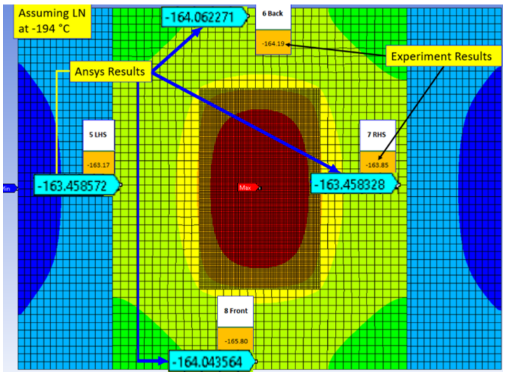

Figure 8: FEA results for the temperature distribution at the top surface of the silicon crystal.

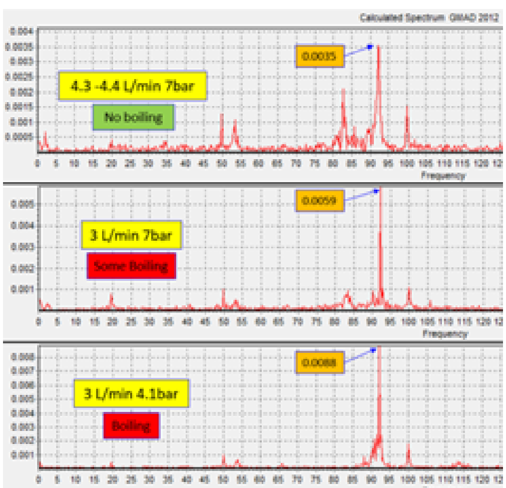

Figure 9: Vibration measurements at different flow rates and working pressures (μm vs. Hz).

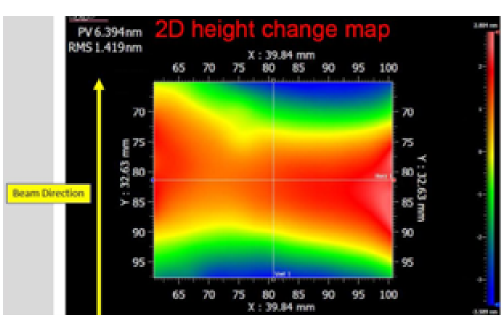

Figure 10: Fizeau interferometer measurements showing the change in the top surface of the crystal due to clamping of cooling endcaps.

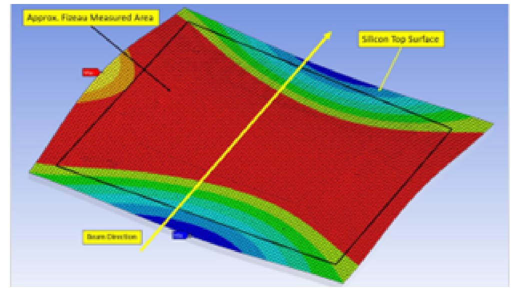

Figure 11: FEA results due to clamping of cooling endcaps.

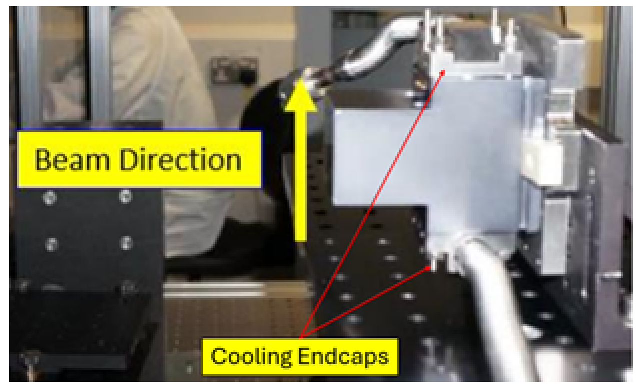

Figure 12: Cooling endcaps.

Table 1: FEA Prediction for Slope Errors

	Total Power (W)	Power Density (W/mm^2)	Flow Rate (L/min)	Slope Errors (μrad)
Si111				
4keV	487	4.26	4.5	±0.43
10keV	600	1.94	4.5	±1.3
19keV	600	1.1	4.5	±0.25
Si311				
8keV	550	28.25	5	±0.7
35keV	700	6.21	5	±0.05
8keV	550	28.25	6	±0.8
35keV	700	6.2	6	±1

CONCLUSION

This study demonstrates that silicon crystal geometry, when directly cooled with LN2, can be tailored to achieve zero slope errors at specified input powers and maintain minimal errors across a wide energy range. Experimental validation confirmed FEA predictions. A sensitivity test increasing flow rate from 5 to 6 L/min raised temperature by 4 °C while slope errors remained below theoretical rocking curve limits. The method provides a practical, efficient way to control thermal deformation in beamline optics, improving resolution and reliability under high heat loads.

ISBN: 978-3-95450-263-9 ISSN: 2673-5520 doi: 10.18429/JACoW-MEDSI2025-WEOA02

REFERENCES

[1] L. Zhang *et al.*, "Thermal deformation of cryogenically cooled silicon crystals under intense X-ray beams: measurement and finite-element predictions of the surface shape", *J. Synchrotron Radiat.*, vol. 20, no. 4, pp. 567–580, May 2013. doi:10.1107/s0909049513009436

[2] A. K. Freund, "Performance of synchrotron x-ray monochromators under heat load: How reliable are the predictions?", in *Proc. SRI'99*, Stanford, CA, USA, Oct. 1999, *AIP Conf. Proc.*, vol. 521, pp. 247–252. doi:10.1063/1.1291795

[3] H. Khosroabadi, L. Alianelli, D. G. Porter, S. Collins, and K. Sawhney, "Cryo-cooled silicon crystal monochromators: a study of power load, temperature and deformation", *J. Synchrotron Radiat.*, vol. 29, no. 2, pp. 377–385, Feb. 2022. doi:10.1107/s160057752200039x